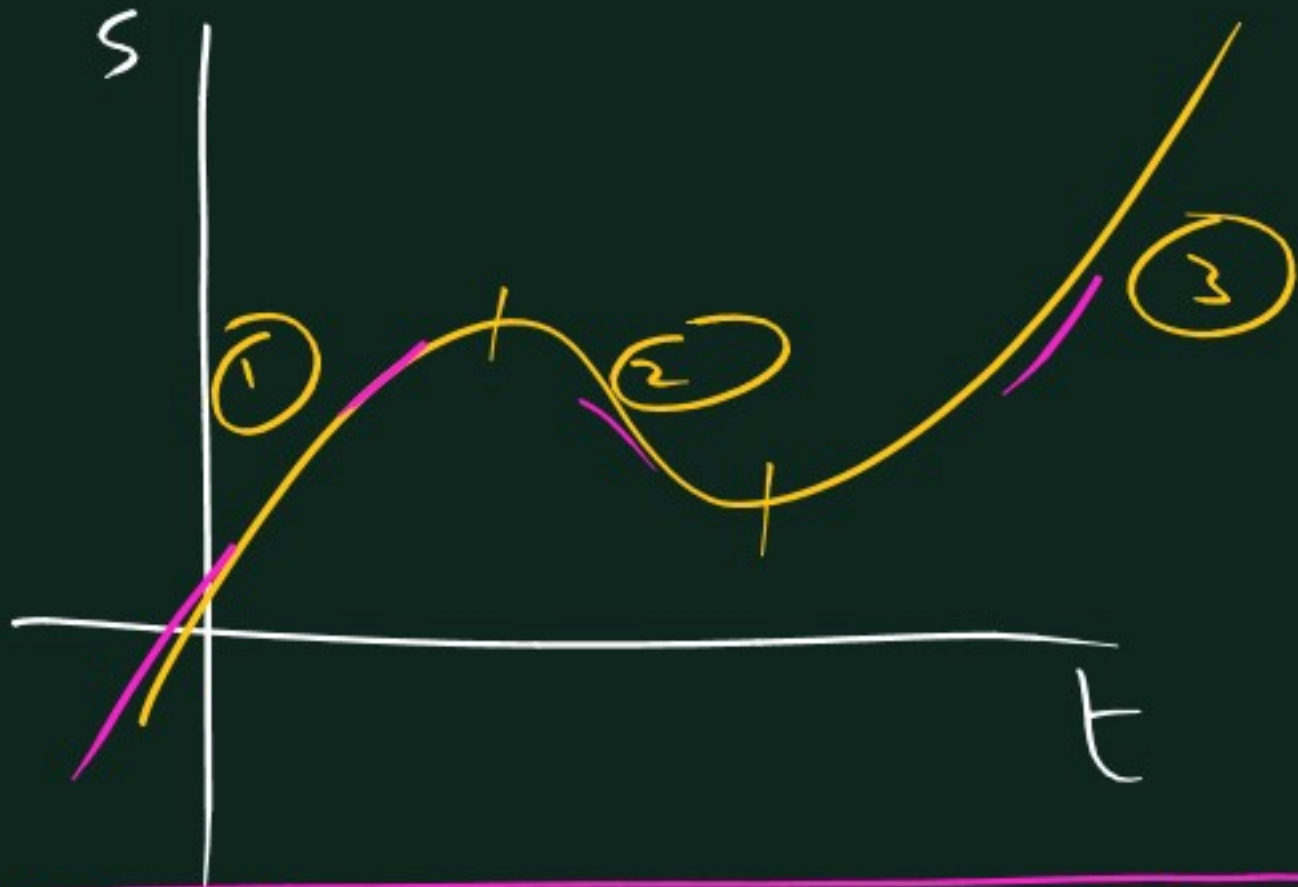


Ex 15)



- ①, ③ → increasing → slope $\frac{ds}{dt} > 0$
 ② → decreasing → " < 0

$$\begin{aligned}
 \frac{ds}{dt} &= 3t^2 - 18t + 24 = \\
 &= 3(t^2 - 6t + 8) \\
 &= \underline{3(t-2)(t-4)}
 \end{aligned}$$

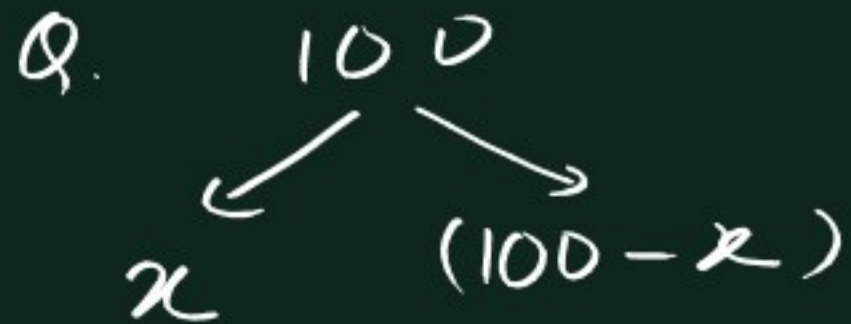
$$\frac{ds}{dt} > 0$$

$$3(t-2)(t-4) > 0$$

$$t < 2 \rightarrow \frac{ds}{dt} (+)$$

$$2 < t < 4 \rightarrow " (-)$$

$$t > 4 \rightarrow " (+)$$



$y = 2x + \underline{(100-x)^2} \leftarrow \underline{\text{minimum}}$

$$y = 2x + 10,000 + x^2 - 200x$$

$$y = x^2 - 198x + 10,000$$

$$\frac{dy}{dx} = 2x - 198 = 0$$

$$\Rightarrow \boxed{x = 99}$$

$$1) \frac{dy}{dx} = 0$$

$$2) \frac{d^2y}{dx^2} > 0 \Rightarrow \text{Min.}$$

$$" < 0 \Rightarrow \text{Max.}$$

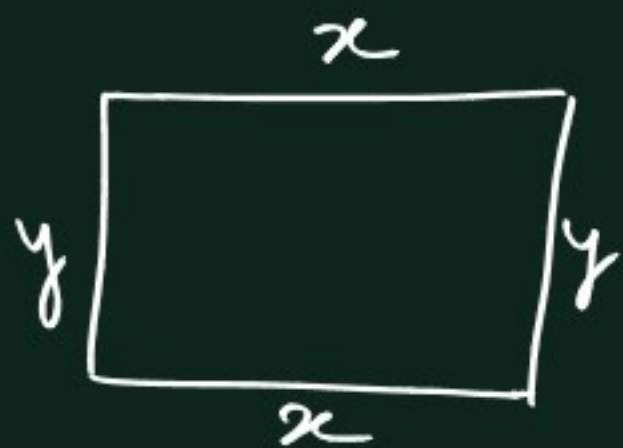
$$\therefore \left. \begin{array}{l} x = 99 \\ 100 - x = 1 \end{array} \right\} \overline{(99, 1)}$$

A) 3

B) 4

C) 5

D) 7



$$\rightarrow \text{Perimeter} = 20$$

$$\rightarrow 2x + 2y = 20$$

$$\Rightarrow x + y = 10$$

$$\Rightarrow \underline{y = (10 - x)}$$



$$D = \sqrt{x^2 + (10 - x)^2}$$

$$\Rightarrow D = \sqrt{2x^2 - 20x + 100}$$

For D to be min. $(2x^2 - 20x + 100)$ will also be min.

$$4x - 20 = 0$$

$$\Rightarrow \boxed{x = 5}$$

$\therefore$ sides $5, 10 - 5$

$$\Rightarrow \underline{(5, 5)}$$

$$\rightarrow \sin x \xrightarrow{D} \cos x$$

I

$$\rightarrow e^x \xrightarrow{D} e^x$$

I

$$\rightarrow \ln x \xrightarrow{D} \frac{1}{x}$$

I

$$\sin x + 2 \xrightarrow{D} \cos x$$

$$\sin x + c \xrightarrow{D} \cos x$$

$\int f(x) dx$

Symbol of I^n

Intⁿ of $f(x)$
w.r.t x

Ex. $\int \cos x dx = \sin x + c$

$$1) \int x^n dx \rightarrow \frac{x^{n+1}}{n+1} + c \quad (n \neq -1)$$

$$\int x^{-1} dx \text{ or } \int \frac{1}{x} dx = \ln x + c$$

$$i) \int x dx = (n=1) \frac{x^{1+1}}{1+1} + c = \frac{x^2}{2} + c$$

$$ii) \int \sqrt{x} dx = \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} = \frac{2}{3} x^{3/2} + c$$

$$iii) \int \frac{1}{x^2} dx = \frac{x^{-2+1}}{-2+1} = \frac{x^{-1}}{-1} = -\frac{1}{x} + c$$

$$iv) \int \underline{dx} = \int x^0 dx = \frac{x^{0+1}}{0+1} = \underline{x + c}$$

$$2) \int \sin x \, dx = -\cos x + c$$

$$\int \cos x \, dx = \sin x + c$$

$$\int \sec^2 x \, dx = \tan x + c$$

$$\int \tan x \sec x \, dx = \sec x + c$$

$$\int \operatorname{cosec}^2 x \, dx \rightarrow -\cot x + c$$

$$\int \cot x \operatorname{cosec} x \, dx \rightarrow -\operatorname{cosec} x + c$$

$$3) \int e^x \, dx = e^x + c$$

$$4) \int (f(x) \pm g(x)) \, dx$$

$$= \int f(x) \, dx \pm \int g(x) \, dx$$

$$\left(\int (x + \sin x) \, dx = \frac{x^2}{2} - \cos x + c \right)$$

$$5) \int k f(x) \, dx = k \int f(x) \, dx$$

$$81. \quad \int \frac{2}{x} dx = 2 \int \frac{1}{x} dx = \underline{2 \ln x + C}$$

$$2) \quad \int (\underline{x^2} - 3x) dx = \frac{x^3}{3} - 3 \frac{x^2}{2} + C$$

$$3) \quad \int (4 \sin x - e^{\frac{x}{2}}) dx = 4(-\cos x) - \frac{1}{\frac{1}{2}} e^{\frac{x}{2}} = \underline{-4 \cos x - \frac{e^x}{2} + C}$$

$$4) \quad \int (3x^2 - 4x + 2) dx = 3 \times \frac{x^3}{3} - 4 \times \frac{x^2}{2} + 2 \boxed{\int dx}^x + C = \underline{x^3 - 2x^2 + 2x + C}$$

$$5) \quad \int \left(\frac{1}{\sqrt{x}} + \frac{4}{x^3} \right) dx = \int x^{-1/2} dx + 4 \int x^{-3} dx = \frac{x^{-1/2+1}}{-1/2+1} + 4 \frac{x^{-3+1}}{(-3+1)} + C$$

$$= 2\sqrt{x} + 4 \frac{x^{-2}}{-2} + C$$

$$= \underline{2\sqrt{x} - \frac{2}{x^2} + C}$$

Substitution rule.

Q. $\int \frac{\cos x \, dx}{\sin^2 x}$ — (1)

Let $\sin x = t$
 $\Downarrow$ Diff. w.r.t t

$$\frac{d(\sin x)}{dx} = \left(\frac{dt}{dx} \right) = 1$$

$$\Rightarrow \frac{d \sin x}{dx} \times \frac{dx}{dt} = 1$$

$$\cos x \frac{dx}{dt} = 1$$

$$\cos x \, dx = dt \quad (2)$$

Put in (1)

$$\int \frac{dt}{t^2}$$

$$= \int t^{-2} dt$$

$$= \frac{t^{-2+1}}{-2+1} + C$$

$$= -\frac{1}{t} + C$$

$$= -\frac{1}{\sin x} + C$$

Q2. $\int \frac{\sec^2 x \, dx}{\sqrt{\tan x}}$

Let $\tan x = t$

$$Q2. \int \frac{\sec^2 x \, dx}{\sqrt{\tan x}}$$

Q3.

$$\int e^{x^2} x \, dx$$

$$Q4. \int \cos^3 x \sin x \, dx$$

$$Q5. \int \sin(3x+1) \, dx$$

$$\text{Let } \tan x = t$$

$$\downarrow$$

$$\frac{d(\tan x)}{dx} \cdot \frac{dx}{dt} = \frac{dt}{dt}$$

$$\Rightarrow \sec^2 x \, dx = dt$$

$$\therefore \int \frac{dt}{\sqrt{t}}$$

$$= \int t^{-1/2} dt$$

$$= \frac{t^{-1/2+1}}{-1/2+1} + C = 2\sqrt{t} + C$$

$$= 2\sqrt{\tan x} + C$$

$$\text{Let } x^2 = t$$

$$\frac{d(x^2)}{dx} \cdot \frac{dx}{dt} = \frac{dt}{dt}$$

$$\Rightarrow 2x \, dx = dt$$

$$\Rightarrow x \, dx = \frac{dt}{2}$$

$$\therefore \int e^t \frac{dt}{2} = \frac{1}{2} \int e^t dt$$

$$= \frac{1}{2} e^t + C = \frac{1}{2} e^{x^2} + C$$

$$\rightarrow \cos x = t$$

$$\frac{d(\cos x)}{dx} \cdot \frac{dx}{dt} = \frac{dt}{dt}$$

$$\Rightarrow -\sin x \, dx = dt$$

$$\Rightarrow \sin x \, dx = -dt$$

$$\therefore \int t^3 (-dt) = -\int t^3 dt$$

$$= -\frac{t^4}{4} + C = -\frac{\cos^4 x}{4} + C$$

$$\# \text{ Let } 3x+1 = t$$

$$3 \, dx = dt$$

$$\Rightarrow dx = \frac{dt}{3}$$

$$\therefore \int \sin t \frac{dt}{3}$$

$$= \frac{1}{3} \int \sin t \, dt$$

$$= -\frac{\cos t}{3} + C$$

$$= -\frac{\cos(3x+1)}{3} + C$$

$$\begin{aligned}
 * \int \sin(ax+b) dx &= -\frac{\cos(ax+b)}{a} + C \\
 \int \cos(ax+b) dx &= \frac{\sin(ax+b)}{a} + C \\
 \int (ax+b)^n dx &= \frac{(ax+b)^{n+1}}{a(n+1)} + C \\
 \int \frac{1}{ax+b} dx &= \frac{\ln(ax+b)}{a} + C \\
 \int e^{(ax+b)} dx &= \frac{e^{ax+b}}{a} + C
 \end{aligned}
 \left. \vphantom{\begin{aligned} * \int \sin(ax+b) dx \\ \int \cos(ax+b) dx \\ \int (ax+b)^n dx \\ \int \frac{1}{ax+b} dx \\ \int e^{(ax+b)} dx \end{aligned}} \right\} \text{Let } \underline{(ax+b)=t}$$

$$\begin{aligned}
 Q1) \int e^{4x+1} dx \\
 = \frac{e^{4x+1}}{4} + C
 \end{aligned}$$

$$\begin{aligned}
 Q2) \int ((2x-1)^4 - \cos(2x)) dx \\
 \frac{(2x-1)^5}{5 \times 2} - \frac{\sin 2x}{2} + C \\
 = \frac{(2x-1)^5}{10} - \frac{\sin 2x}{2} + C
 \end{aligned}$$

Definite Integration

$$Q1) \int_0^2 x \, dx$$

$$\underline{\underline{Ans}} \quad \left| \frac{x^2}{2} \right|_0^2$$

$$= \frac{1}{2} \left| x^2 \right|_0^2$$

$$= \frac{1}{2} |2^2 - 0^2|$$

$$= \frac{1}{2} \times 4 = \boxed{2}$$

$$\int_a^b f(x) \, dx$$

upper limit $\rightarrow b$

lower limit $\rightarrow a$

- 1) Integrate
 - 2) Put limits & Subtract

$\rightarrow$ No C

$\rightarrow$ Some numerical answer.

$$Q2. \int_0^{\pi/2} \cos x \, dx$$

$$= \left| \sin x \right|_0^{\pi/2}$$

$$= \left| \sin \frac{\pi}{2} - \sin 0 \right|$$

$$= 1 - 0$$

$$= 1$$

$$Q3. \int_0^2 (4x^2 - \underline{2}) \, dx$$

$$= \left| 4 \frac{x^3}{3} - 2x \right|_0^2$$

$$= \left(4 \frac{2^3}{3} - 2 \times 2 \right) - 0$$

$$= \boxed{20/3}$$

$$Q4. \int_{-1}^2 x^2 \, dx$$

$$\int_{-1}^2 x^2 dx$$

$$= \left| \frac{x^3}{3} \right|_{-1}^2$$

$$= \frac{1}{3} \left| x^3 \right|_{-1}^2$$

$$= \frac{1}{3} \left| (2^3) - \underbrace{(-1)^3}_{-1} \right|$$

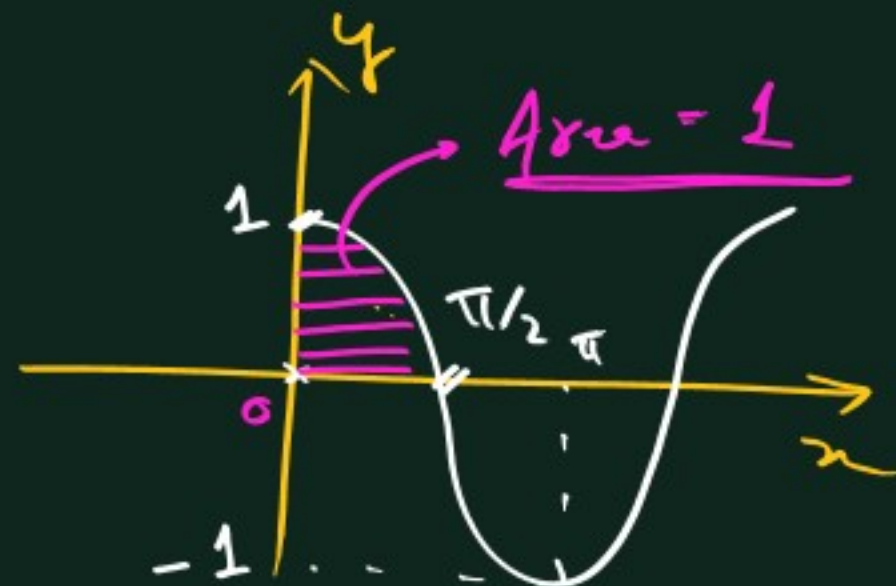
$$= \frac{1}{3} \left| 8 - (-1) \right|$$

$$= \textcircled{3}$$

$$\int_a^b f(x) dx$$

$$y = f(x)$$

$$\int_0^{\frac{\pi}{2}} \cos x dx = 1$$

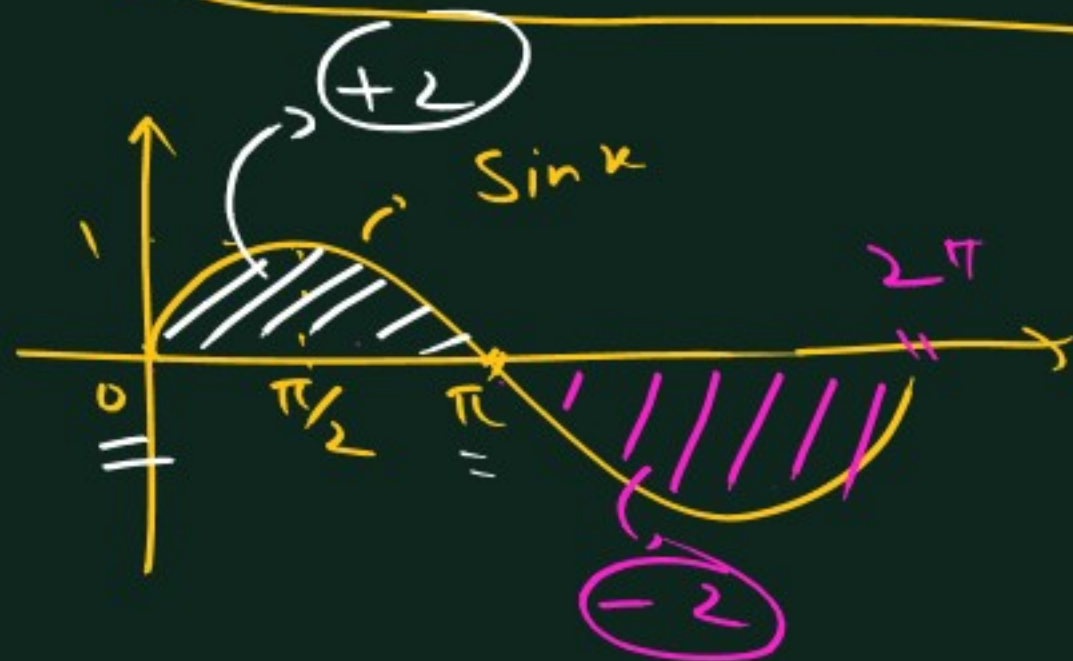


$$\cos 2\pi = 1$$

$$\sin 2\pi = 0$$

(Eg.) $y = \sin x$

$$\int_0^{\pi} \sin x dx$$



$$= -\cos x \Big|_0^{\pi}$$

$$= -(\cos \pi - \cos 0) = -(-1 - 1) = -(-2) = +2$$

Also find

$$\int_0^{2\pi} \sin x dx$$

$$= -\cos x \Big|_0^{2\pi}$$

$$= -(\cos 2\pi - \cos 0)$$

$$= -(1 - 1) = 0$$

$$y = (x^2 + 1)$$



$$\int_0^2 (x^2 + 1) dx$$

$$= \left| \frac{x^3}{3} + x \right|_0^2$$

$$= \left| \frac{8}{3} + 2 \right| - 0$$

$$= \frac{14}{3}$$

$$1) \quad v = \frac{dx}{dt} \text{ or } \frac{ds}{dt}$$

(x = position / s = disp.)

$$2) \quad a = \frac{dv}{dt} \text{ or } a = \frac{d^2x}{dt^2}$$

$$3) \quad a = \frac{dv}{dt} = \frac{dv}{ds} \times \frac{ds}{dt}$$

$$a = v \frac{dv}{ds}$$

$$Q1. \quad x = t^3 + 2t \quad t=0$$

Find 1) initial v & a

2) v, a at $t = 2s$.

3) Nature of acceleration
(uniform/non-uniform)

Ans 1) $v = \frac{dx}{dt} = 3t^2 + 2$

$$a = \frac{dv}{dt} = 6t$$

$t=0$ (initial) $\Rightarrow$ $v = 2 \text{ m/s}$
 $a = 0$

$$2) \quad t = 2s$$

$$v = 3 \times 2^2 + 2 = 14 \text{ m/s}$$

$$a = 6t = 6 \times 2 = 12 \text{ m/s}^2$$

$$3) \quad a = 6t$$

$$t = 0, a = 0$$

$$t = 1, a = 6$$

$$t = 2, a = 12$$

non-uniform acceleration.

Q2. $x = 3t^2 - 4t + 1$

Find time when $a = v$

Aus: $v = 6t - 4$

$a = \frac{dv}{dt} = 6$

$\therefore a = v$

$\Rightarrow 6 = 6t - 4$

$\Rightarrow \boxed{t = 5/3}$

Q3. Coordinates of a particle are $x = 2t^2$, $y = 3t^2$

Find 1) $\vec{v}$

2) speed at $t = 2$ sec.

$\left(\frac{dx}{dt} = v_x, \frac{dy}{dt} = v_y \right)$

$\vec{v} = v_x \hat{i} + v_y \hat{j}$

1) $\frac{dx}{dt} = 4t, \frac{dy}{dt} = 6t$

$\vec{v} = 4t \hat{i} + 6t \hat{j}$

2) $t = 2$
 $\vec{v} = 8\hat{i} + 12\hat{j}$

Speed = $|\vec{v}| = \sqrt{8^2 + 12^2} = \underline{\underline{4\sqrt{13}}}$

Q4. $v = x^2$

Find a at

$x = 2 \text{ m}$

$\left(a = v \frac{dv}{dx} \text{ if } v \rightarrow f(x) \right)$

$a = v \left(\frac{dv}{dx} \right) \rightarrow \frac{d(x^2)}{dx}$
 $\downarrow$
 $= 2x$

$\Rightarrow a = x^2 \times 2x$
 $= \underline{\underline{2x^3}}$

$x = 2$, $a = 2 \times 2^3$
 $= \underline{\underline{16 \text{ m/s}^2}}$

If $\frac{dy}{dx} = f(x)$

Ex.

$\frac{dy}{dx} = x^2$

If $y=1$ at $x=0$,
find y at any x

Ans

$\int_{y=1}^y dy = \int_{x=0}^x x^2 dx$

$a = 6t$

$|y|_1^4 = |x|_0^3$

$\Rightarrow y^4 - 1 = x^3 - 0$

$\Rightarrow y = x^{\frac{3}{4}} + 1$

$v = (x^2 + 1)$

1) To find position x from v

$\frac{dx}{dt} = v(t)$

$\Rightarrow \int dx = \int v dt$

2) To find v from a

$\frac{dv}{dt} = a \Rightarrow \int dv = \int a dt$

3) $a = v \frac{dv}{dx}$

$\Rightarrow \int a dx = \int v dv$

1) $a = 6t$
 $t=0 \rightarrow x=10$
 $u=0$ (starts from rest)

$$\frac{dv}{dt} = 6t$$

$$\int dv = 6 \int t dt$$

$$v=0 \quad t=0$$

$$\left| v \right|_0^t = \left| 3t^2 \right|_0^t$$

$$\Rightarrow v-0 = 3(t^2-0)$$

$$\Rightarrow v = 3t^2$$

$$\frac{dx}{dt} = 3t^2$$

$$\Rightarrow \int dx = 3 \int t^2 dt$$

$$x=10 \quad t=0$$

$$\Rightarrow \left| x \right|_{10}^t = \left| \frac{3t^3}{3} \right|_0^t$$

$$\Rightarrow x-10 = t^3 \Rightarrow \boxed{x = t^3 + 10}$$

Q2. $v = (3t+4)$

Find displacement

in $t = 2$ sec.

$$\frac{ds}{dt} = (3t+4) \Rightarrow \int ds = \int (3t+4) dt$$

3) $a = 2 - \frac{t}{10}$

For car starting with 10 m/s. find 'v' and displacement in $t=10$ sec.

(At $t=0$, displacement will be 0)

$$\Rightarrow S \Big|_0^2 = \left(3t^2 + 4t \right) \Big|_0^2$$

$$\Rightarrow S = \left(3 \times \frac{4}{2} + 4 \times 2 \right) - 0$$

$$\Rightarrow \boxed{S = 14m}$$

$$3) a = 2 - \frac{t}{10}$$

$$(u = 10 \text{ m/s})$$

$$\frac{dv}{dt} = \left(2 - \frac{t}{10}\right)$$

$$\Rightarrow \int_{u=10}^v dv = \int_{t=0}^{10} \left(2 - \frac{t}{10}\right) dt$$

$$\Rightarrow \left|v\right|_{10}^v = \left|2t - \frac{t^2}{20}\right|_0^{10}$$

$$\Rightarrow v - 10 = \left(2 \times 10 - \frac{100}{20}\right) - 0$$

$$\Rightarrow v - 10 = 15$$

$$\Rightarrow \boxed{v = 25}$$

To find Disp. find $v(t)$

$$\int_{10}^v dv = \int_0^t \left(2 - \frac{t}{10}\right) dt$$

$$\Rightarrow \left|v\right|_{10}^v = \left|2t - \frac{t^2}{20}\right|_0^t$$

$$\Rightarrow v - 10 = 2t - \frac{t^2}{20} \Rightarrow \boxed{v = 2t - \frac{t^2}{20} + 10}$$

$$\text{Put } v = \frac{ds}{dt}$$

$$\Rightarrow \frac{ds}{dt} = \left(2t - \frac{t^2}{20} + 10\right)$$

$$\Rightarrow \int ds = \int_0^{10} \left(2t - \frac{t^2}{20} + 10\right) dt$$

$$= \left| \frac{2t^2}{2} - \frac{t^3}{60} + 10t \right|_0^{10}$$

$$= \left| 100 - \frac{1000}{60} + 100 \right| - 0$$

$$= 200 - \frac{50}{3}$$

$$\boxed{s = \frac{550}{3} = 183.3 \text{ m}}$$

Q4) $v = 2t^2 + \sin t$

At $t=0$ particle is at origin. Find

1) position x & a at time t

2) initial speed & a

① $a = \frac{dv}{dt} = 4t + \cos t$

For x $\frac{dx}{dt} = 2t^2 + \sin t$

$\Rightarrow \int_{x=0}^x dx = \int_0^t (2t^2 + \sin t) dt$

$\Rightarrow \left| x \right|_0^x = \left| \frac{2}{3}t^3 - \cos t \right|_0^t \Rightarrow x - 0 = \left(\frac{2}{3}t^3 - \cos t \right) - (0 - \cos 0) = \frac{2}{3}t^3 - \cos t + 1$

5) given $a = x^2$

1) Find v as $f(x)$ if

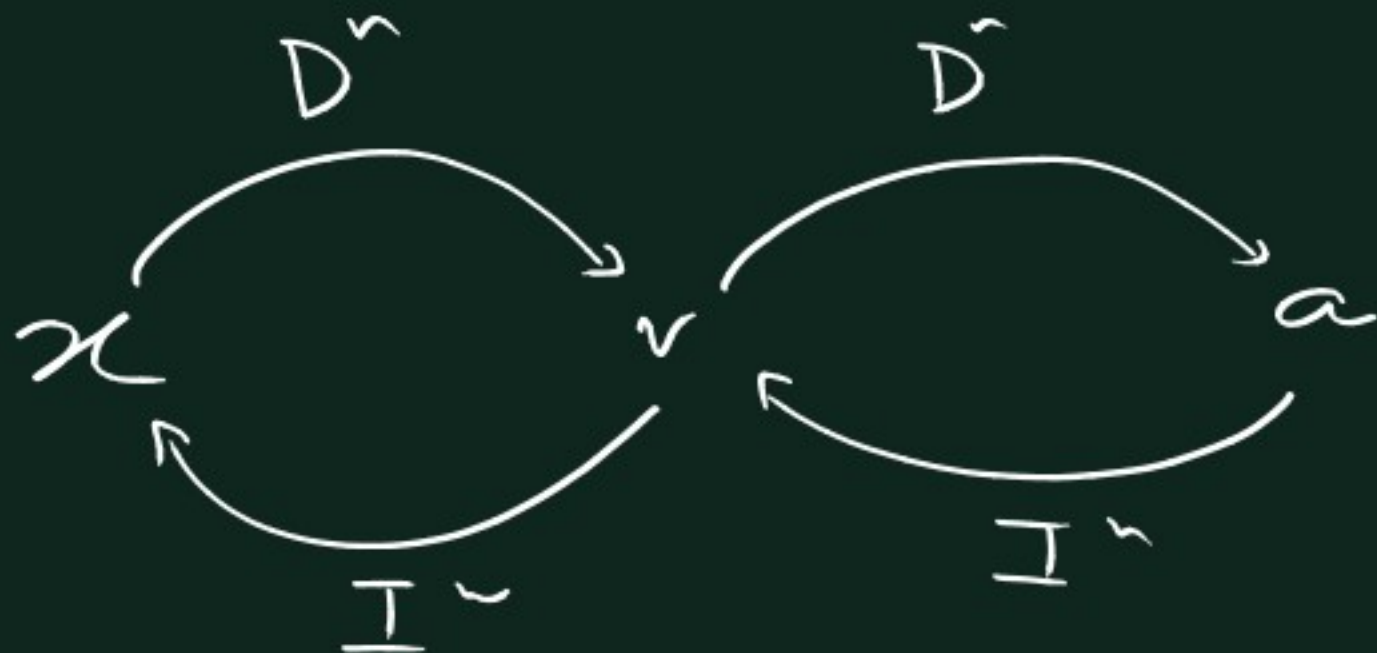
$v=0$ at $x=0$

2) x at time t if

$x=1$ at $t=0$

2) $\underline{t=0}$ $a = 0 + \cos 0 = 1$

$v = 0 + \sin 0 = 0$



$$a = x^2$$

1) $v(x) = ??$ if $x=0, v=0$

2) $x(t)$ if $t=0, x=1$

$$v \frac{dv}{dx} = x^2$$

$$\Rightarrow \int_0^x v dv = \int_0^x x^2 dx$$

$$\Rightarrow \left| \frac{v^2}{2} \right|_0^x = \left| \frac{x^3}{3} \right|_0^x$$

$$\Rightarrow \frac{v^2}{2} = \frac{x^3}{3} \Rightarrow v = \sqrt{\frac{2x^3}{3}}$$

$$2) \quad \boxed{v = \sqrt{\frac{2x^3}{3}}} \quad A) \sqrt{6t+1}$$

$$\Rightarrow \frac{dx}{dt} = \sqrt{\frac{2}{3}} x^{3/2}$$

$$\int_1^x \frac{dx}{x^{3/2}} = \sqrt{\frac{2}{3}} \int_0^t dt \quad \left(\int x^{-3/2} dx = \frac{x^{-3/2+1}}{-3/2+1} \right)$$

$$\Rightarrow \left| \frac{x^{-1/2}}{-1/2} \right|_1^x = \sqrt{\frac{2}{3}} t$$

$$\Rightarrow -2 \left| \frac{1}{\sqrt{x}} - \frac{1}{\sqrt{1}} \right| = \sqrt{\frac{2}{3}} t$$

$$\Rightarrow \frac{1}{\sqrt{x}} - 1 = \left(\sqrt{\frac{2}{3}} t \right)$$

$$\Rightarrow \frac{1}{\sqrt{x}} = 1 + \frac{t}{\sqrt{6}}$$

$$\Rightarrow \frac{1}{x} = \left(1 + \frac{t}{\sqrt{6}} \right)^2$$

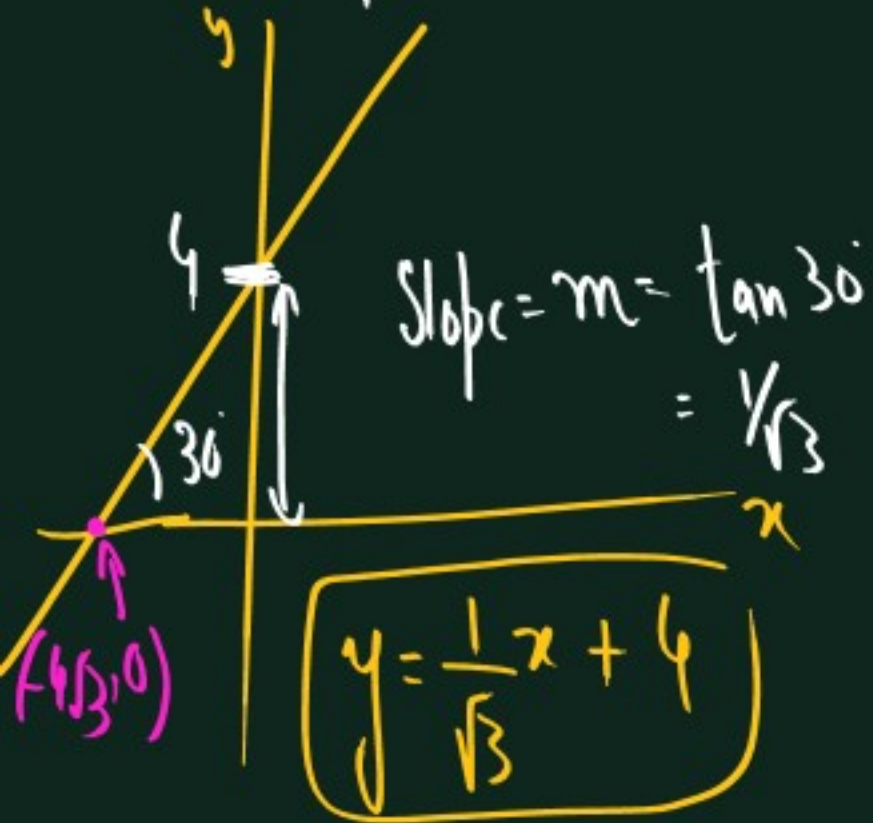
$$\Rightarrow \boxed{x = \frac{1}{\left(1 + \frac{t}{\sqrt{6}} \right)^2}}$$

Some basic graphs.

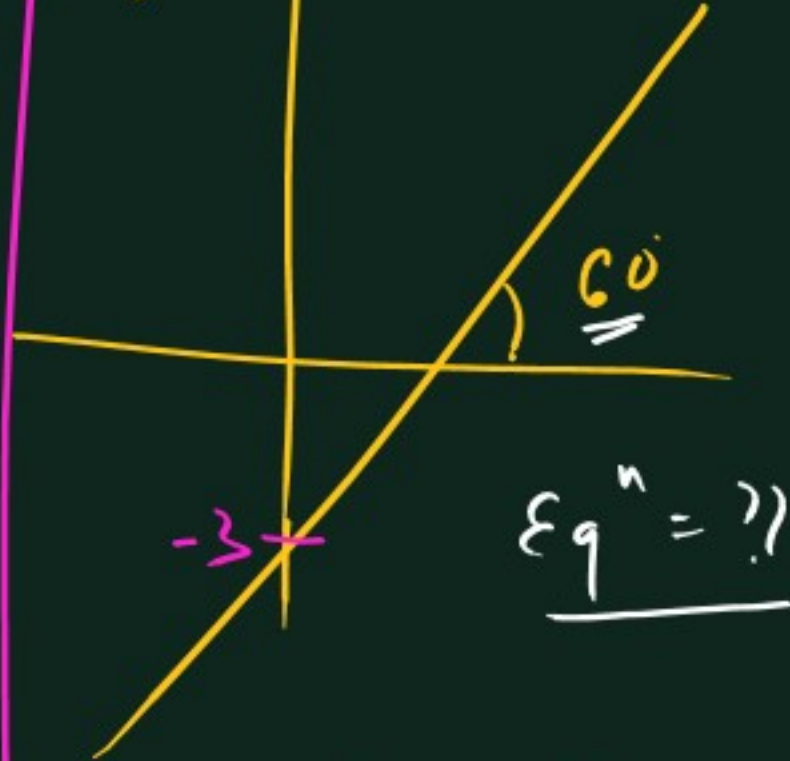
1) Straight line

$$y = \underbrace{m}_\text{Slope} x + \underbrace{c}_\text{y-intercept}$$

$$\text{Slope} = \tan \theta$$



ξ.

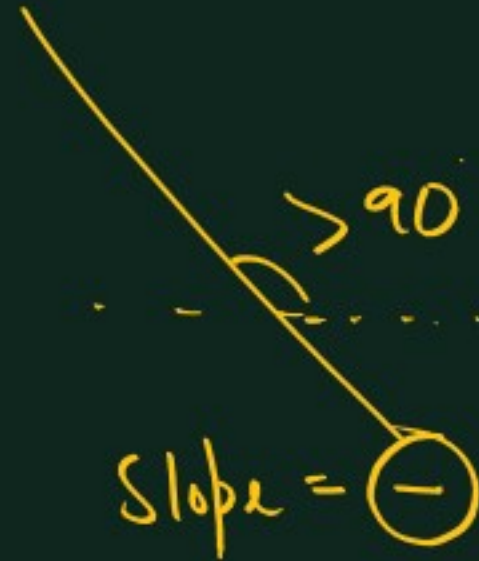
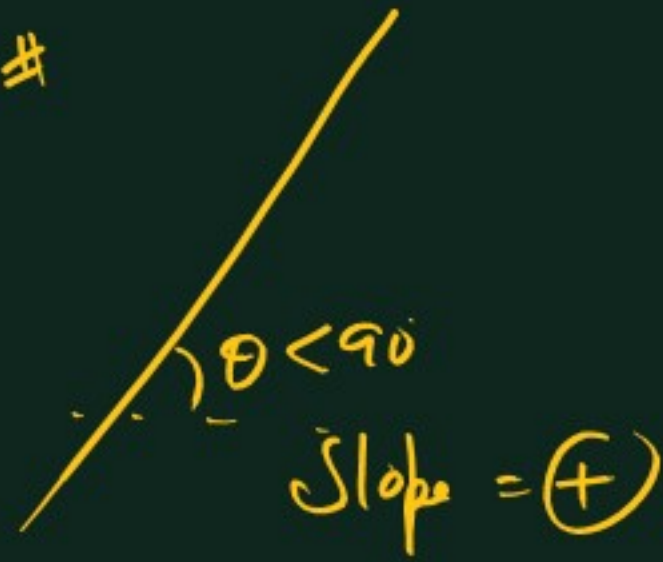


$$y = \underbrace{m}_\text{Slope} x + \underbrace{c}_\text{y-intercept}$$

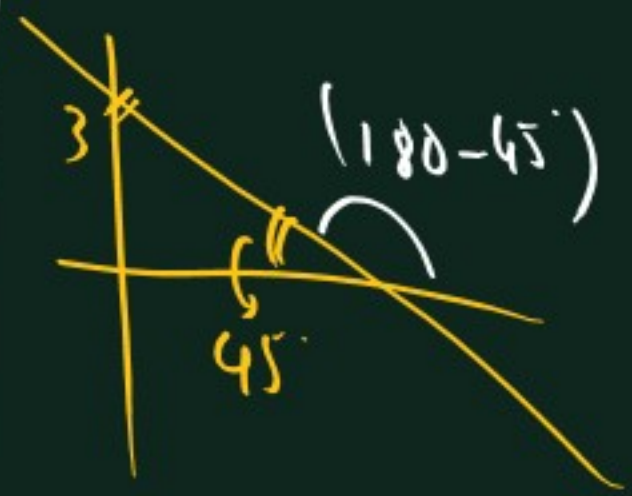
$$\tan 60^\circ = \sqrt{3}$$

$$\Rightarrow \boxed{y = \sqrt{3}x - 3}$$

#



$$\tan(180^\circ - \theta) = -\tan \theta$$



$$y = \underbrace{m}_\text{Slope} x + \underbrace{c}_\text{y-intercept}$$

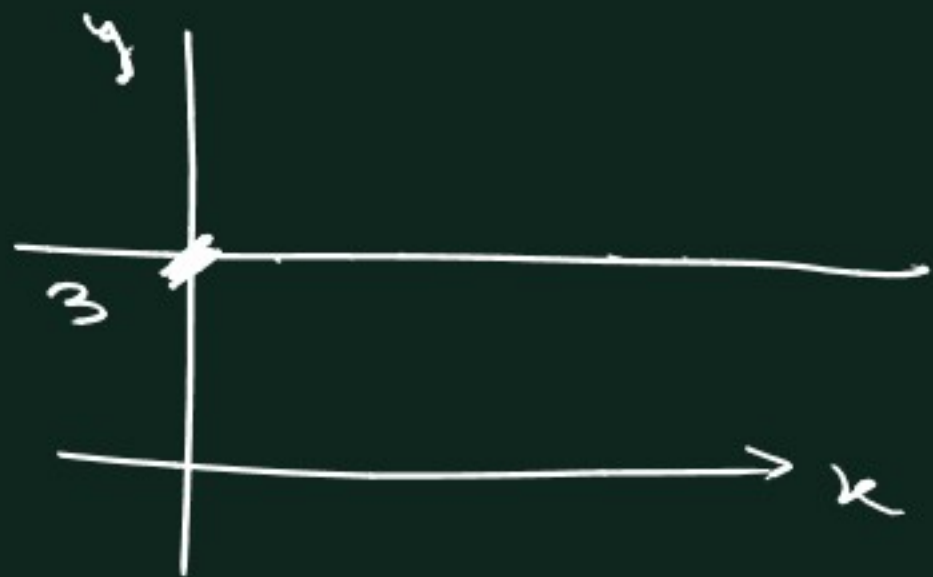
$$\tan(180^\circ - 45^\circ)$$

$$= -\tan 45^\circ$$

$$= -1$$

$$\Rightarrow \boxed{y = -x + 3}$$

$$y = c$$



$$y = 3$$

$$y = \underline{ax^2 + bx + c}$$



$$x = ay^2 + by + c$$



$$xy = c$$

$$\text{or } y \propto \frac{1}{x}$$

